

PPOM2- OMM MINI-LECTURE: Introduction to Articulatory (ART) and Myofascial Release (MFR) Techniques

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Objectives

1. Students will understand the principles of ART and MFR and their applications.
2. Students will understand the indications and contraindications of ART and MFR.
3. Students will understand the basic technique principles of ART and MFR.
4. Students will understand the principles of regional vs. segmental vs. peripheral ART and their applications.
5. Students will understand the models of MFR and their applications.

ART Definition:

- Articulatory method is a direct, joint focused group of techniques, which use low-velocity, high-amplitude (LVHA) movements. In performing treatment, the physician gently carries the body region being treated through the restrictive barrier. This may be done as a single movement or repetitively. The repetitive techniques are often referred to as springing. Many articulatory techniques use long lever forces to achieve their effects.

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ART Indications:

- Restricted motion in the presence of articular and/or myofascial somatic dysfunction (especially in the frail or elderly)
- Circulatory and lymphatic congestion

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ART Contraindications:

- Acute moderate to severe strain or sprain
- Fracture, dislocation, or joint instability in the area affected by the treatment
- Acute inflammatory joint disease in the area affected by the treatment
- Metastasis in the area affected by the treatment
- Neurologic entrapment syndromes
- Serious vascular compromise
- Infection (e.g., cellulitis, abscess, septic arthritis, osteomyelitis)
- Bleeding disorders

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ART Basic Technique Principles:

1. The patient should be comfortable and able to relax.
2. The physician should also be in a position of comfort to minimize energy expenditure and to make use of body weight and leverage rather than having to exert muscular force.
3. The physician moves the affected joint or body part through its range of motion until the restrictive barrier is engaged.
4. A gentle but firm force is applied, carrying the body region a short distance through the restrictive barrier. The distance is limited by the tissues being treated and the patient's tolerance to pain or fatigue.
5. This force is typically applied rhythmically, consisting of 1 or 2 seconds of stretch followed by a similar time frame releasing the stretch. The joint is permitted to return to a point just short of its restrictive barrier.
6. As the patient responds to the technique, the restrictive barrier will shift position within the physiologic range of motion. For each cycle of the applied technique, re-engage the restrictive barrier and carry the affected body region a short distance farther, through the new barrier to normal motion.
7. The applied forces should be comfortable to the patient. Some patients experience discomfort, but it is recognized by the patient as a good and necessary discomfort.
8. The technique is continued until the location of the restrictive barrier reaches a plateau; that is, no further increase in range of motion can be achieved by continuing the technique or until full physiologic range of motion has been restored to the joint(s) being treated.
9. At no time is the anatomic barrier to joint motion exceeded.

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ART Types:

- Regional: ART may be applied to the entire regions of the spine to improve regional range of motion, when the restriction to motion is perceived by the physician to have an articular component to it.
Ex: Cervical Sidebending
- Segmental: ART are applied to segmental motion restrictions that are perceived to be articular in nature in the vertebral column.
Ex: Segmental Somatic Dysfunction—C4
- Peripheral: ART are especially useful in treating restriction of motion of the peripheral limb joints.

Ex: Radiocarpal Joint Restriction

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ART Key Concepts:

- involve gentle and repetitive motions through the restrictive barrier.
- are one of the most versatile treatment forms available since the physician can use a wide range of force levels and adapt them to a variety of patient body types and locations of dysfunction.
- many techniques use long lever forces in achieving effects.

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MFR Definition:

- is a system of diagnosis and treatment first described by Andrew Taylor Still and his early students, which engages continual palpatory feedback to achieve release of myofascial tissues. It can be applied as a *direct or indirect technique*.
- in a direct technique, a restrictive barrier is identified in the myofascial tissues, then engaged with a loaded, constant, directional force until the tissue releases and motion is restored.
- in an indirect technique, the tissues' position of ease is identified, then engaged with directed pressure. This guides the tissues along the line of least resistance until free movement of all tissues is achieved.
- a *combined procedure*, whereby the physician's hands engage both a barrier and point of ease simultaneously, allows the MFR to occur through the application of both direct and indirect methods interactively.

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Activating forces used in MFR techniques:

- A. *Inherent (intrinsic) force* uses the body's natural tendency to seek homeostasis. Inherent force is the rhythmic activity in all tissues, which works to improve the hydrodynamic and bioenergetic factors around restricted tissues and articulations.
- B. *Respiratory force (cooperation/assist)* may be used in four ways:
1. While using a direct or indirect position, *full cycle respiratory effort* is applied as a fascial and/or articulatory activating force.
 2. Once positioned, a *particular phase of respiration* enhances the position of the area being treated.
 3. *Breath holding* for the maximum time that a patient can tolerate causes air hunger and triggers a generalized relaxation of soft tissues.
 4. *Coughing or sniffing on command* produces a respiratory impulse to assist in the release of restrictions.

Activating forces used in MFR techniques (cont'd):

- C. *Patient cooperation*: The patient is requested to move in specific directions in various planes of an articulation, to aid in mobilizing an area of restriction.
- D. *Physician-guided force*: After engaging a barrier or point of ease, the physician sequentially guides the tissue or joint through various positions, following a shifting pattern of easy motion, until the path of dysfunction is retraced and released.
- E. *Springing/vibration*: Placing the hands or percussion hammer on a dysfunction, and applying variable degrees of pressure and/or frequency of force causing springing or vibration in the structure, activates release of the tissues.

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Models of MFR Techniques

Models of MFR	Techniques	Tissue Focus	Emphasizes
Biomechanical model	MFR, integrated INR	Muscular–fascial relationships	Mechanics, anatomic relationships, tight/loose concepts, tethering, and interdependent neural relationships
Fascial continuum of the big bandage model	Fascial–ligamentous release, fascial release	Fascia	Fascial continuum, treat proximal to distal, fulcrum with leverage, torsion and traction, respiratory assistance, and reflexive neurologic reset
Bioenergetic model	Bioelectric fascial activation and release, fascial release, MFR	Fascia, holographic interface in homeostasis	Bioresponsive electrical potentials, diaphragms, longitudinal cables and tubes, fascial interface in homeostasis, holographic diagnosing and treating

MFR Indications:

- to treat somatic dysfunction involving myofascial tissues or other connective tissues
- to alleviate the visceral and somatic components of somatic dysfunction in the connective tissues of the body, with a focus on fascia, muscles, and related structures (neuroreflexive, circulatory, lymphatic, respiratory, bioenergetic, systemic, holographic, and homeostatic).
- to address structural imbalances and somatic dysfunction indicative of and pursuant to changes in homeostasis in systemic disease.
- to aid in the normalization of function in all systems.

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MFR techniques address somatic dysfunction by:

- Normalizing motion
- Relieving edema
- Reestablishing symmetry
- Relieving pain
- Aiding circulatory and lymphatic function
- Normalizing neuroreflexive activity
- Supporting visceral function
- Restoring bioenergetic balance
- Supporting homeostatic function

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MFR Contraindications:

Cautions and relative contraindications for local application:

- Fractures
- Open wounds
- Soft tissue or bony infections
- Abscesses
- DVT
- Coagulopathy
- Neoplasm
- Directly over the site of recent operation
- Aortic aneurysm

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MFR General Considerations and Rules:

1. The physician palpates the patient using layer-by-layer palpatory principles and with just enough pressure to capture the skin and subcutaneous fascial structures. This is one level of pressure less than that of soft tissue technique. Any movement of the hand on the skin should cause the skin to move along with the hand without sliding the hand over the skin.
2. The physician gently moves the palpating hand or hands in a linear direction of choice (hands of the clock) moving through the x- and y-axes. The z-axis has already been engaged by the layer-by-layer palpatory pressure into the body, gaining access to the superficial fascia.
3. Symmetry versus asymmetry of tissue compliance is noted in the linear directions tested.
4. The physician may add a variety of directions of motion, including other linear movements in a 360-degree reference and clockwise and counterclockwise rotational movement. Again, symmetry versus asymmetry of tissue compliance is noted.
5. The pressure the physician uses to determine compliance may be minimal or moderate, depending on the clinical presentation of the patient (acute painful vs. chronic minimally painful) and what the physician believes is appropriate for the situation.

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MFR General Considerations & Rules (cont'd):

6. After determining the ease and bind barriers of the tissue in these directions, the physician determines whether gentle or moderate pressure in a direct (toward bind) or indirect (toward ease) technique is appropriate. Again, this is determined by the clinical presentation and examination findings. In general, the gentlest method is the safest.
7. The physician slowly moves the hand-controlled myofascial tissues toward the appropriate barrier, and on meeting the barrier, he or she holds the tissue at that point without relieving the pressure. The physician should notice that after approximately 20 to 30 seconds, a change of tissue compliance occurs; this is demonstrated by movement of the tissue through the originally determined barrier (creep or fascial creep).
8. The physician follows this change and continues to engage the barrier until no further evidence of creep occurs. There may be a number of compliance changes (creep) before this phenomenon stops.
9. The physician reevaluates the tissue to determine whether the tissue's compliance and quality have improved. The technique may be repeated at the same area or another, and follow-up visits may be prescribed for a 3-day interval or longer, depending on patient reactivity.

MFR Key Concepts:

- emphasizes appreciation and recruitment of the innate and inherent motion of the myofascial structures.
- identifies patterns of motion and ease across proximal and distal structures, diaphragms, compartments, viscera, and organ systems.
- includes myofascial release (MFR), integrated neuromusculoskeletal release, fascial–ligamentous release, and bioelectric fascial activation and release.
- respiratory assistance, fulcrums, body positions, neuroreflexive activities, oscillations, passive progressive motions, and muscular contractions may aid in the progression of MFR.

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